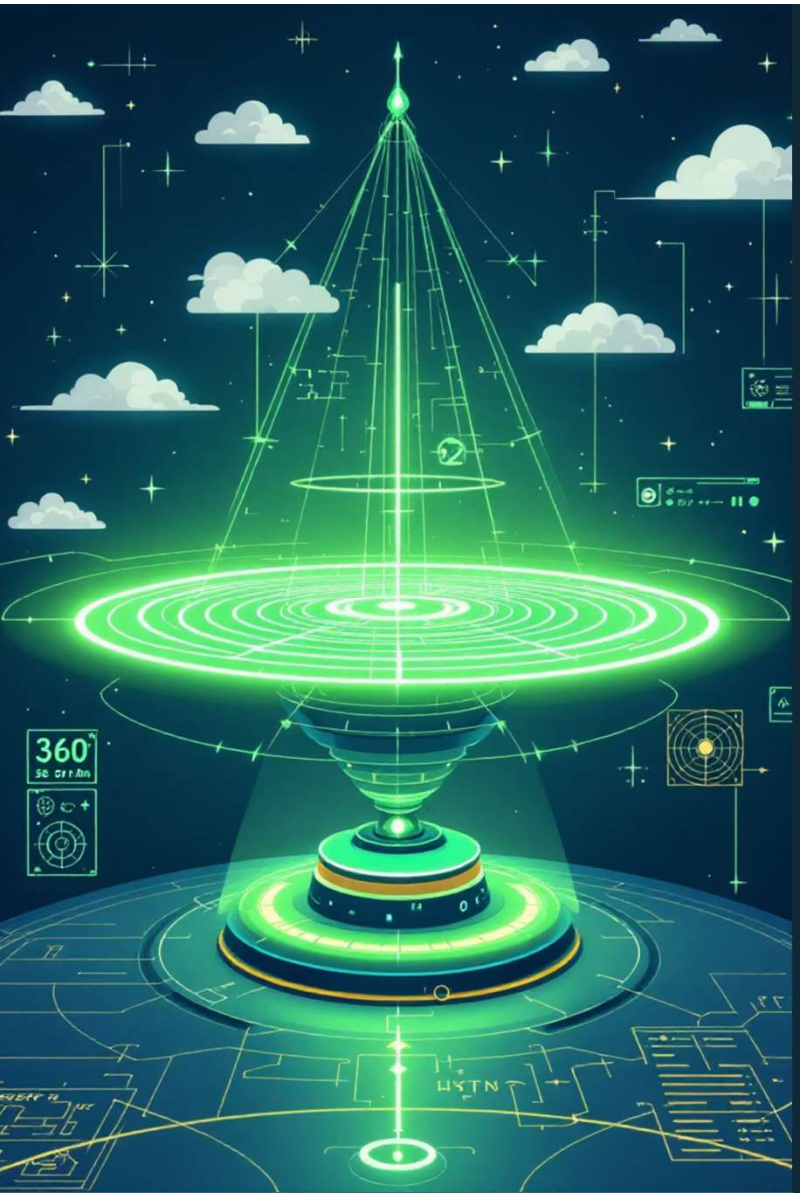


360-Degree Ultrasonic Object Sensor

Implementation using HC-SR04 Sensor and Servo Motor

Project Overview

Objective

Design a low-cost radar system for 360° detection.

Function

Visualizes distance and angle data in real-time.

Why 360°?

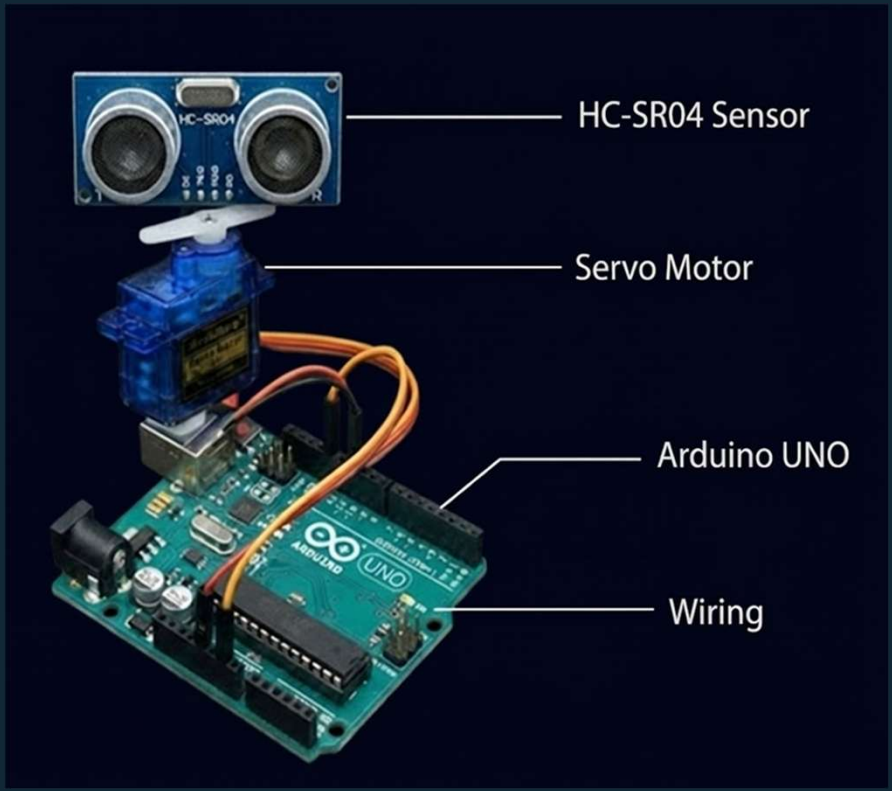
Eliminates blind spots standard sensors cannot see.

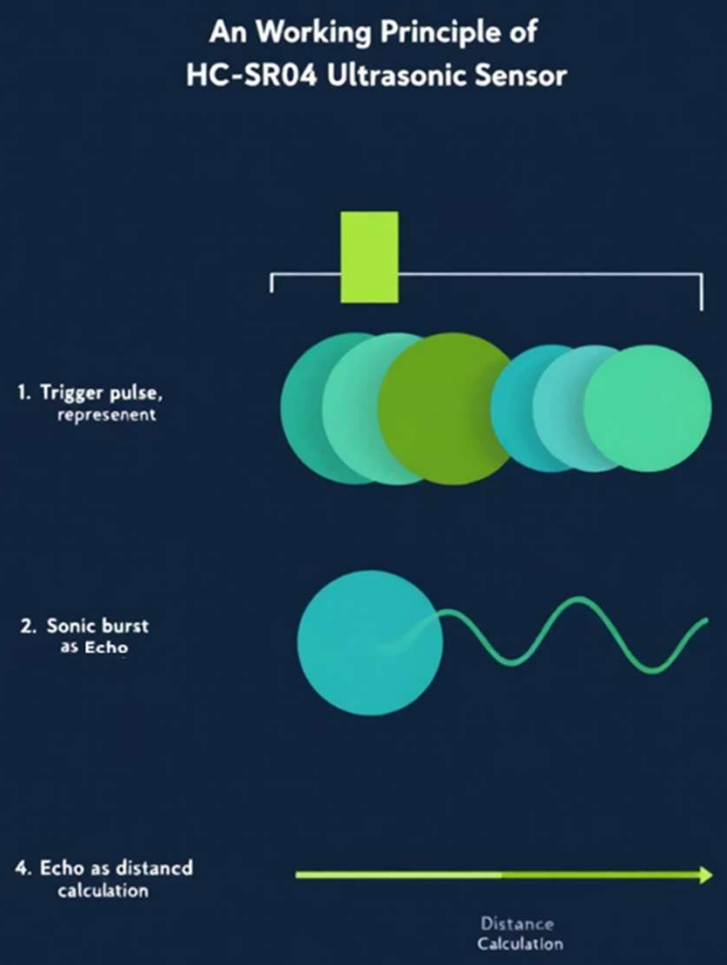
Key Benefit

Provides situational awareness for autonomous robots.

Hardware Components

- **HC-SR04 Ultrasonic Sensor:** The "eyes" emitting 40kHz sound waves.
- **Servo Motor (SG90):** Rotates sensor 0° to 180° for scanning.
- **Arduino UNO:** Microcontroller that triggers sensor and controls servo.
- **Wiring:** Jumper wires and breadboard setup.





Working Principle of HC-SR04

01

Trigger Pulse

Arduino sends 10µs High pulse.

02

Sonic Burst

Sensor emits 8 cycles of ultrasonic sound.

03

Echo

Sound reflects off object and returns to sensor.

04

The Formula

Distance = (Speed of Sound × Time) / 2.

Servo Rotation & Coverage

- **Sweeping Motion:** Servo scans incrementally (e.g., 0° to 180°).
- **Step Size:** Pauses every 1 degree for stable reading.
- **Data Mapping:** Creates polar coordinates (Angle, Distance).
- **Logic:** Move Servo → Stop → Measure Distance → Repeat.

Measurement & Data Flow

1

Step 1

Servo aligns to specific Angle.

2

Step 2

HC-SR04 measures Distance.

3

Step 3

Arduino sends string "Angle,Distance" via Serial Port.

4

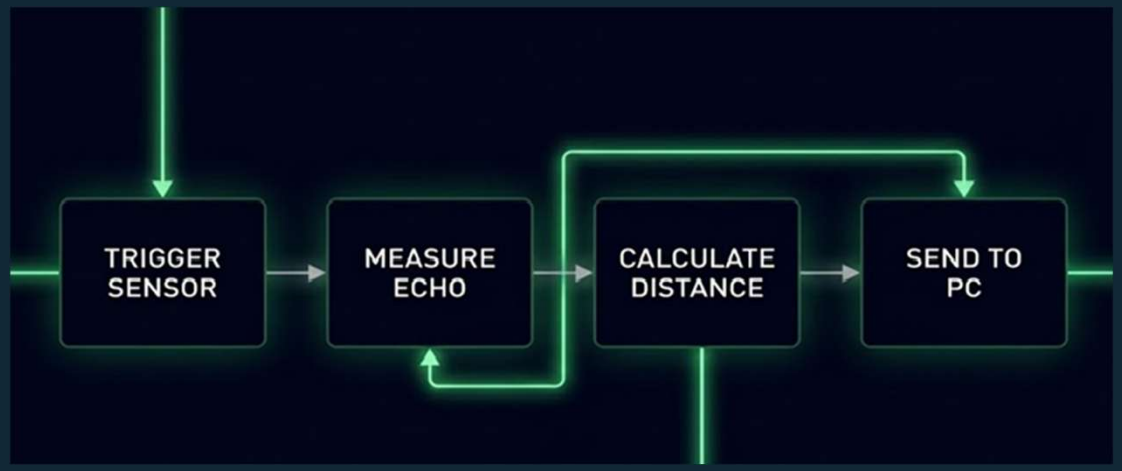
Step 4

PC Software (Processing IDE) parses data.

5

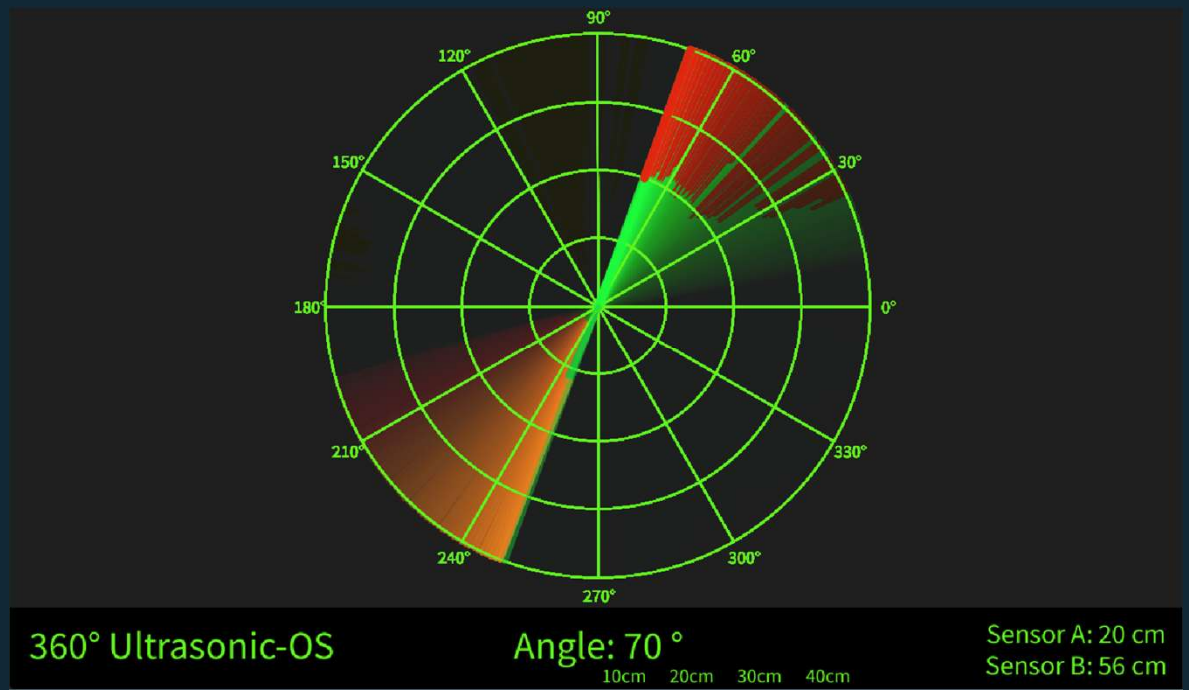
Step 5

Point is plotted on the visual interface.



Radar-Style Visualization (The UI)

- **Sweeping Line:** Green line mimics the physical servo arm.
- **Fade Effect:** Old detection points fade to keep map current.
- **Color Coding:** Red dots for close range (danger), Green for safe.
- **Polar Grid:** Concentric circles showing distance (10cm, 20cm,30cm).



Applications

Robotics

Obstacle avoidance and SLAM (Simultaneous Localization and Mapping).

Security

Intruder detection in defined zones.

Automotive

Parking assist and proximity sensing.

Industrial

Conveyor belt monitoring and object counting.

Limitations

Dead Zone

Cannot detect objects closer than 2cm.

Soft Surfaces

Fabric or sponge absorbs sound (no echo).

Reflection Angle

Sharp angles deflect sound away from the sensor.

Speed

Sound is slower than light, limiting scan speed.

Future Improvements

1

LiDAR Integration

Using laser for millimeter accuracy.

2

Continuous Rotation

Using slip-rings for full 360° spinning.

3

Noise Filtering

Kalman filters to smooth sensor jitter.

4

3D Mapping

Adding a vertical servo to create point clouds.

References

[1] A. S. Morris, *Measurement and Instrumentation: Theory and Application*. Boston, MA, USA: Academic Press, 2011.

[2] J. L. León-Rodríguez, J. A. Pérez-Quiroz, and J. Escobedo-Alatorre, “Ultrasonic distance measurement system using HC-SR04 sensor for educational robotics,” in *Proc. IEEE Int. Conf. Mechatronics*, 2015, pp. 1–6.

[3] T. R. Ayodele and O. O. Ogundele, “Development of a low-cost ultrasonic-based obstacle detection system,” *IEEE Trans. Instrum. Meas.*, vol. 68, no. 8, pp. 3147–3155, Aug. 2019.

[4] A. K. Singh, P. Gupta, and R. Kumar, “Design and implementation of servo-controlled scanning ultrasonic radar,” in *Proc. IEEE Int. Conf. Robotics and Automation Sciences*, 2017, pp. 312–316.

[5] Texas Instruments, “Ultrasonic sensing basics,” *TI Application Report*, SLYY187, 2018.

[6] S. F. Barrett and D. J. Pack, *Atmel AVR Microcontroller Primer: Programming and Interfacing*. New York, NY, USA: Springer, 2012.

[7] S. Choi and J. Kim, “Real-time visualization techniques for radar-like UI systems,” *IEEE Access*, vol. 7, pp. 115400–115412, 2019.

THANK
YOU